

Surface Mount Frequency Mixer

Level 7 (LO Power +7 dBm) 5 to 1500 MHz

ADE-5+



Generic photo used for illustration purposes only

CASE STYLE: CD542

+RoHS Compliant

The +Suffix identifies RoHS Compliance. See our web site for RoHS Compliance methodologies and qualifications

Maximum Ratings

Operating Temperature	-40°C to 85°C
Storage Temperature	-55°C to 100°C
RF Power	50mW
IF Current	40mA

Permanent damage may occur if any of these limits are exceeded.

Pin Connections

LO	6
RF	3
IF	2
GROUND	1,4,5

Features

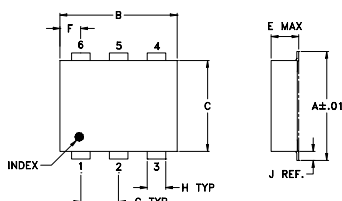
- low conversion loss, 6.6 dB typ.
- excellent L-R isolation, 40 dB typ.
- excellent IP3, 15 dBm
- aqueous washable
- protected by U.S. Patent 6,133,525

Applications

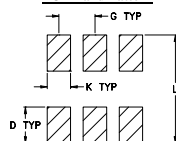
- cellular
- PCS

Available Tape and Reel at no extra cost	
Reel Size	Devices/Reel
7"	20, 50, 100, 200
13"	500, 1000

Outline Drawing



PCB Land Pattern

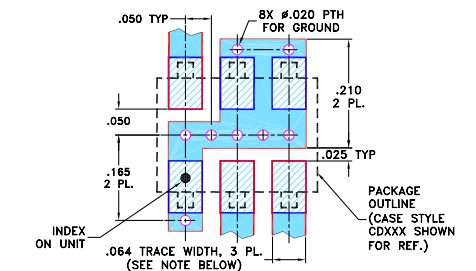


Suggested Layout,
Tolerance to be within ±.002

Outline Dimensions (inch)

A	B	C	D	E	F	G
.272	.310	.220	.100	.112	.055	.100
6.91	7.87	5.59	2.54	2.84	1.40	2.54
H	J	K	L	wt		
.030	.026	.065	.300	grams		
0.76	0.66	1.65	7.62	0.20		

Demo Board MCL P/N: TB-03 Suggested PCB Layout (PL-052)



- NOTES: 1. TRACE WIDTH IS SHOWN FOR ROGERS RO4350B WITH DIELECTRIC THICKNESS .030" ± .002"; COPPER: 1/2 OZ. EACH SIDE. FOR OTHER MATERIALS TRACE WIDTH MAY NEED TO BE MODIFIED.
2. BOTTOM SIDE OF THE PCB IS CONTINUOUS GROUND PLANE.

- DENOTES PCB COPPER LAYOUT WITH SMOBC (SOLDER MASK OVER BARE COPPER)
- DENOTES COPPER LAND PATTERN FREE OF SOLDER MASK

Electrical Specifications

FREQUENCY (MHz)		CONVERSION LOSS (dB)				LO-RF ISOLATION (dB)			LO-IF ISOLATION (dB)			IP3 at center band (dBm)						
LO/RF f _L -f _U	IF	Mid-Band m		Total Range Max.		L	M	U	L	M	U							
		$\overline{X}$	σ	Max.									Typ. Min.	Typ. Min.	Typ. Min.	Typ. Min.	Typ. Min.	Typ. Min.
5-1500	DC-1000	6.6	0.10	7.5	9.3	50	40	40	25	33	23	50	40	30	20	20	10	15

1 dB COMP.: +1 dBm typ.

Phase detection, positive polarity

L = low range [f_L to $10 f_L$]

m = mid band [$2f_L$ to $f_U/2$]

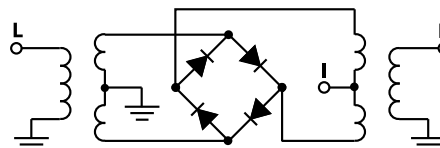
M = mid range [$10 f_L$ to $f_U/2$]

U = upper range [$f_U/2$ to f_U]

Typical Performance Data

Frequency (MHz)		Conversion Loss (dB)	Isolation L-R (dB)	Isolation L-I (dB)	VSWR RF Port (:1)	VSWR LO Port (:1)
RF	LO	LO +7dBm	LO +7dBm	LO +7dBm	LO +7dBm	LO +7dBm
5.0	35.0	7.0	63.9	53.6	1.59	1.91
10.0	40.0	6.6	63.2	52.2	1.51	1.94
50.0	80.0	6.5	56.7	46.6	1.39	1.91
100.0	130.0	6.6	52.7	41.7	1.38	1.91
197.1	227.1	6.7	49.0	36.3	1.41	1.94
295.6	325.6	6.6	47.3	33.5	1.40	2.01
394.1	424.1	6.7	43.7	32.0	1.39	2.20
492.6	522.6	6.5	42.4	30.7	1.35	2.40
500.0	530.0	6.6	41.9	30.5	1.34	2.37
591.1	621.1	6.6	39.7	29.9	1.35	2.40
689.6	719.6	6.6	36.8	28.5	1.28	2.32
788.1	818.1	6.6	35.2	26.0	1.23	2.30
886.6	916.6	6.5	33.4	23.5	1.14	2.32
985.1	1015.1	6.4	33.2	21.8	1.09	2.49
1000.0	1030.0	6.5	33.5	21.5	1.11	2.58
1182.0	1212.0	6.6	35.6	20.4	1.50	2.88
1280.5	1310.5	7.1	36.3	19.2	1.81	2.92
1379.0	1409.0	7.6	34.3	17.9	2.10	2.92
1477.5	1507.5	8.2	31.2	17.1	2.37	2.88
1500.0	1530.0	8.2	30.5	16.8	2.46	2.96

Electrical Schematic



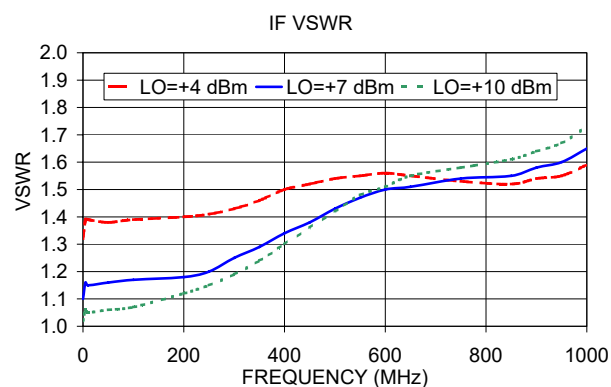
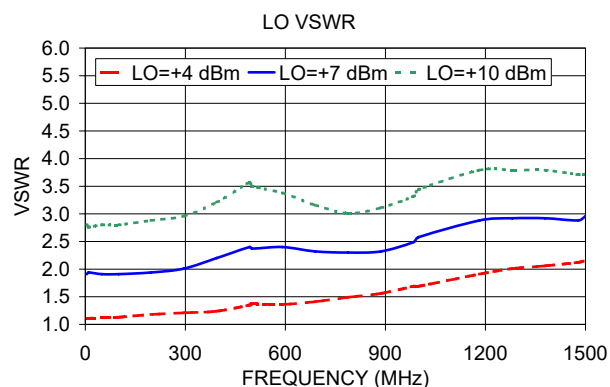
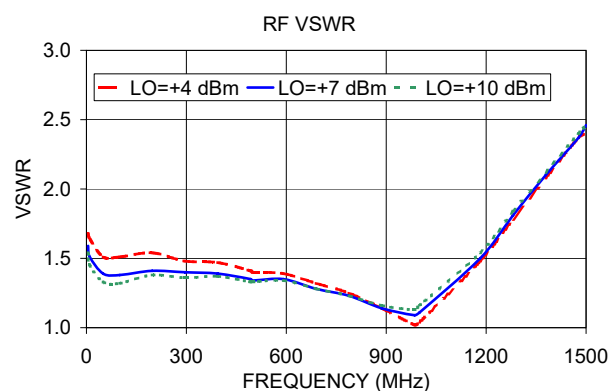
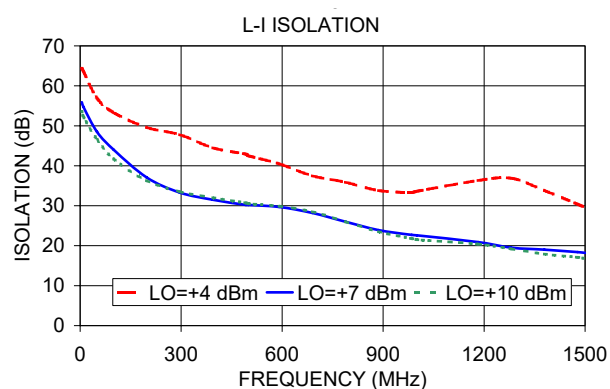
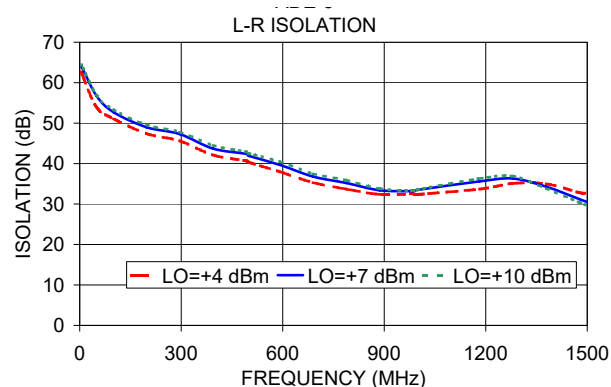
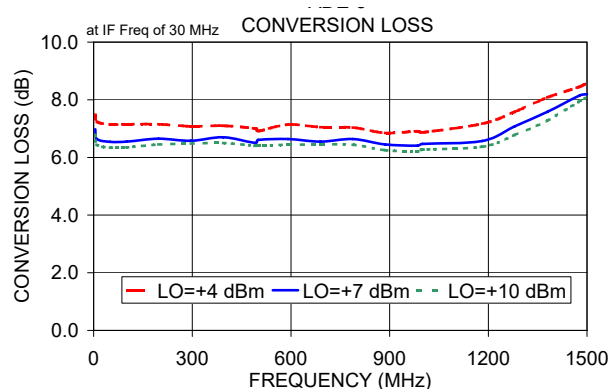
Notes

- A. Performance and quality attributes and conditions not expressly stated in this specification document are intended to be excluded and do not form a part of this specification document.
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Typical Performance Data

RF (IN) (MHz)	LO (MHz)	CONVERSION LOSS IF FIXED @IF(OUT)=30MHz (dB)			RF (IN) (MHz)	LO (MHz)	IP3 INPUT (dBm)			RF (IN) (MHz)	LO (MHz)	COMPRESSION @RF IN=+1dBm (dB)		
		@LO (dBm)					@LO (dBm)					@LO (dBm)		
+4	+7	+10	+4	+7	+10	+4	+7	+10	+4	+7	+10			
5.0	35.0	7.48	6.98	6.78	10.1	40.1	15.58	20.51	23.05	10.1	40.1	0.45	0.31	0.25
10.0	40.0	7.24	6.64	6.44	49.9	79.9	15.26	20.08	20.07	49.9	79.9	0.53	0.33	0.28
49.9	79.9	7.37	6.84	6.60	89.7	119.7	14.88	18.96	20.00	89.7	119.7	0.50	0.41	0.28
89.7	119.7	7.41	6.89	6.65	129.5	159.5	15.49	15.95	20.95	129.5	159.5	0.49	0.40	0.23
129.5	159.5	7.42	6.87	6.66	169.3	199.3	14.96	17.56	20.45	169.3	199.3	0.49	0.41	0.26
169.3	199.3	7.36	6.88	6.66	209.0	239.0	14.80	17.96	19.47	209.0	239.0	0.57	0.38	0.24
209.0	239.0	7.33	6.87	6.69	248.8	278.8	14.87	18.74	23.99	248.8	278.8	0.55	0.34	0.32
248.8	278.8	7.31	6.89	6.69	308.5	338.5	14.65	18.93	17.31	308.5	338.5	0.47	0.39	0.23
308.5	338.5	7.34	6.94	6.75	348.3	378.3	17.66	18.72	17.98	348.3	378.3	0.60	0.37	0.28
348.3	378.3	7.33	6.95	6.74	408.0	438.0	15.65	17.13	18.08	408.0	438.0	0.62	0.36	0.27
408.0	438.0	7.37	6.98	6.78	447.8	477.8	16.68	20.38	17.17	447.8	477.8	0.50	0.39	0.28
447.8	477.8	7.40	7.00	6.80	507.5	537.5	17.82	17.23	17.43	507.5	537.5	0.49	0.38	0.29
507.5	537.5	7.44	7.04	6.87	547.3	577.3	16.90	17.00	16.06	547.3	577.3	0.46	0.50	0.29
547.3	577.3	7.49	7.08	6.89	606.9	636.9	15.01	15.88	15.85	606.9	636.9	0.52	0.51	0.34
606.9	636.9	7.55	7.10	6.83	646.7	676.7	15.61	15.91	15.01	646.7	676.7	0.70	0.65	0.55
646.7	676.7	7.58	7.17	6.89	706.4	736.4	16.16	16.86	17.45	706.4	736.4	0.63	0.54	0.46
706.4	736.4	7.66	7.24	6.99	746.2	776.2	16.41	17.01	16.57	746.2	776.2	0.70	0.57	0.52
805.9	835.9	7.68	7.24	6.94	805.9	835.9	15.03	15.52	14.82	805.9	835.9	0.83	0.73	0.58
845.7	875.7	7.67	7.17	6.89	845.7	875.7	13.72	14.23	14.92	845.7	875.7	1.08	0.93	0.76
905.4	935.4	7.73	7.18	6.86	905.4	935.4	12.88	12.10	13.20	905.4	935.4	1.15	1.07	0.93
945.2	975.2	7.80	7.19	6.85	945.2	975.2	12.26	11.50	12.30	945.2	975.2	1.17	1.16	0.96
1004.8	1034.8	7.88	7.26	6.89	1004.8	1034.8	11.09	11.31	11.03	1004.8	1034.8	1.27	1.16	0.97
1044.6	1074.6	7.91	7.28	6.89	1044.6	1074.6	10.10	10.80	12.00	1044.6	1074.6	1.10	1.01	0.97
1104.3	1134.3	7.96	7.36	6.94	1104.3	1134.3	8.96	9.96	11.18	1104.3	1134.3	1.18	1.08	0.96
1144.1	1174.1	8.00	7.42	7.00	1144.1	1174.1	8.35	9.98	10.73	1144.1	1174.1	1.19	1.09	1.04
1203.8	1233.8	8.02	7.51	7.12	1203.8	1233.8	7.74	9.22	9.32	1203.8	1233.8	1.30	1.07	0.99
1243.6	1273.6	8.04	7.54	7.22	1243.6	1273.6	7.33	8.33	10.20	1243.6	1273.6	1.33	1.05	0.96
1303.3	1333.3	7.97	7.50	7.16	1303.3	1333.3	8.15	10.60	12.48	1303.3	1333.3	1.48	1.16	1.00
1343.0	1373.0	7.97	7.52	7.21	1343.0	1373.0	8.92	11.15	12.65	1343.0	1373.0	1.47	1.22	1.01
1402.7	1432.7	8.00	7.56	7.29	1402.7	1432.7	9.92	12.80	13.65	1402.7	1432.7	1.38	1.11	0.88
1442.5	1472.5	8.03	7.63	7.37	1442.5	1472.5	10.21	13.09	14.38	1442.5	1472.5	1.32	1.08	0.91
1502.2	1532.2	8.21	7.84	7.66	1502.2	1532.2	10.87	12.83	14.40	1502.2	1532.2	1.28	0.94	0.90
1542.0	1572.0	8.36	8.07	7.88	1542.0	1572.0	11.83	14.09	15.96	1542.0	1572.0	1.19	0.89	0.74
1601.7	1631.7	8.69	8.43	8.29	1601.7	1631.7	11.25	13.16	16.08	1601.7	1631.7	0.95	0.71	0.81
1641.5	1671.5	8.84	8.68	8.59	1641.5	1671.5	11.08	13.06	15.88	1641.5	1671.5	0.74	0.70	0.66
1701.2	1731.2	9.06	8.89	8.82	1701.2	1731.2	11.38	13.94	15.06	1701.2	1731.2	0.72	0.58	0.45
1740.9	1770.9	9.19	9.03	8.97	1740.9	1770.9	12.51	13.39	15.86	1740.9	1770.9	0.87	0.55	0.46
1800.6	1830.6	9.49	9.31	9.24	1800.6	1830.6	13.91	16.85	18.66	1800.6	1830.6	0.69	0.44	0.31
1840.4	1870.4	9.78	9.54	9.44	1840.4	1870.4	15.41	16.40	18.41	1840.4	1870.4	0.51	0.40	0.42
1900.1	1930.1	10.18	9.93	9.87	1900.1	1930.1	14.49	18.96	17.05	1900.1	1930.1	0.61	0.37	0.29

Typical Performance Data

IF (OUT) (MHz)	LO (MHz)	CONVERSION LOSS VS. IF FREQUENCY @RF(IN)=750.1MHz (dB)	IF (OUT) (MHz)	LO (MHz)	CONVERSION LOSS VS. IF FREQUENCY @RF(IN)=10.1MHz (dB)	IF (OUT) (MHz)	LO (MHz)	CONVERSION LOSS VS. IF FREQUENCY @RF(IN)=1500.1MHz (dB)
		@LO (dBm)			@LO (dBm)			@LO (dBm)
		+7			+7			+7
740.0	10.1	7.47	10.0	20.1	6.66	1000.0	500.1	8.51
721.3	28.8	7.48	30.0	40.1	6.24	979.8	520.3	8.33
702.6	47.5	7.46	50.0	60.1	6.16	959.6	540.5	8.26
683.8	66.3	7.47	70.0	80.1	6.24	939.4	560.7	8.20
665.1	85.0	7.40	90.0	100.1	6.22	919.2	580.9	8.15
646.4	103.7	7.37	110.0	120.1	6.12	899.0	601.1	8.06
627.7	122.4	7.32	130.0	140.1	6.16	878.8	621.3	8.03
609.0	141.1	7.25	150.0	160.1	6.20	858.6	641.5	7.97
590.3	159.8	7.23	170.0	180.1	6.18	838.4	661.7	7.95
571.5	178.6	7.19	190.0	200.1	6.21	818.2	681.9	7.95
552.8	197.3	7.18	210.0	220.1	6.20	798.0	702.1	7.92
534.1	216.0	7.14	230.0	240.1	6.28	777.8	722.3	7.90
515.4	234.7	7.12	250.0	260.1	6.22	757.6	742.5	7.87
496.7	253.4	7.10	270.0	280.1	6.27	737.3	762.8	7.88
477.9	272.2	7.09	290.0	300.1	6.36	717.1	783.0	7.84
459.2	290.9	7.08	310.0	320.1	6.35	696.9	803.2	7.79
440.5	309.6	7.11	330.0	340.1	6.37	676.7	823.4	7.73
421.8	328.3	7.09	350.0	360.1	6.44	656.5	843.6	7.72
403.1	347.0	7.05	370.0	380.1	6.41	636.3	863.8	7.66
384.4	365.7	7.04	390.0	400.1	6.46	616.1	884.0	7.63
365.6	384.5	7.01	430.0	440.1	6.86	575.7	924.4	7.53
346.9	403.2	7.04	450.0	460.1	6.63	555.5	944.6	7.54
328.2	421.9	7.04	490.0	500.1	6.59	515.1	985.0	7.50
309.5	440.6	7.02	510.0	520.1	6.64	494.9	1005.2	7.50
290.8	459.3	7.01	550.0	560.1	6.73	454.5	1045.6	7.52
272.1	478.0	7.03	570.0	580.1	6.79	434.3	1065.8	7.59
253.3	496.8	7.04	610.0	620.1	6.77	393.9	1106.2	7.55
234.6	515.5	6.98	630.0	640.1	6.97	373.7	1126.4	7.63
215.9	534.2	7.06	670.0	680.1	6.98	333.3	1166.8	7.69
197.2	552.9	7.04	690.0	700.1	7.05	313.1	1187.0	7.74
178.5	571.6	7.04	730.0	740.1	7.19	272.7	1227.4	7.84
159.7	590.4	7.03	750.0	760.1	7.28	252.4	1247.7	7.89
141.0	609.1	7.04	790.0	800.1	7.24	212.0	1288.1	7.96
122.3	627.8	7.06	810.0	820.1	7.24	191.8	1308.3	8.01
103.6	646.5	7.10	850.0	860.1	7.23	151.4	1348.7	7.99
84.9	665.2	7.12	870.0	880.1	7.13	131.2	1368.9	7.96
66.2	683.9	7.14	910.0	920.1	7.17	90.8	1409.3	7.94
47.4	702.7	7.15	930.0	940.1	7.24	70.6	1429.5	7.91
28.7	721.4	7.26	970.0	980.1	7.30	30.2	1469.9	7.81
10.0	740.1	7.14	990.0	1000.1	7.30	10.0	1490.1	7.77

Typical Performance Data

LO (MHz)	LO-RF ISOLATION (dB)			LO-IF ISOLATION (dB)			RF (IN) (MHz)	LO (MHz)	RF-IF ISOLATION (dB)		
	@LO (dBm)			@LO (dBm)					@LO (dBm)		
	+4	+7	+10	+4	+7	+10			+4	+7	+10
5.0	62.6	63.9	64.4	55.9	53.6	52.4	10.1	40.1	42.19	49.31	48.33
10.0	61.4	63.2	63.6	54.7	52.2	51.4	49.9	79.9	34.91	34.84	34.69
49.9	61.08	60.98	60.65	55.75	54.61	55.38	89.7	119.7	29.97	30.08	30.18
89.7	55.41	55.60	55.58	49.88	49.37	48.68	129.5	159.5	27.31	27.39	27.37
129.5	52.30	52.23	52.56	45.64	44.95	44.63	169.3	199.3	25.29	25.46	25.52
169.3	50.10	50.23	50.70	42.62	41.75	42.08	209.0	239.0	23.91	24.05	24.11
209.0	48.36	48.80	49.12	39.72	39.51	40.26	248.8	278.8	22.84	23.01	23.11
248.8	46.94	47.56	47.90	37.50	37.92	38.88	308.5	338.5	21.72	21.97	22.10
308.5	45.40	45.98	46.35	35.25	36.14	37.21	348.3	378.3	21.20	21.49	21.67
348.3	44.61	45.30	45.71	34.05	35.20	36.31	408.0	438.0	20.59	20.97	21.21
408.0	43.15	43.93	44.43	32.74	34.07	35.15	447.8	477.8	20.20	20.60	20.87
447.8	42.36	43.14	43.68	31.86	33.18	34.18	507.5	537.5	19.52	19.96	20.26
507.5	41.41	42.42	43.10	31.08	32.51	33.50	547.3	577.3	19.12	19.65	20.03
547.3	40.59	41.65	42.39	30.48	31.87	32.81	606.9	636.9	18.85	19.54	20.02
606.9	39.04	40.18	41.02	29.97	31.20	31.87	646.7	676.7	18.66	19.33	19.84
646.7	37.90	39.08	39.97	29.70	30.95	31.53	706.4	736.4	18.13	18.63	18.98
706.4	36.33	37.46	38.35	29.10	30.30	30.82	746.2	776.2	17.48	17.80	18.00
805.9	34.21	35.20	35.96	27.33	28.08	28.44	805.9	835.9	16.34	16.43	16.46
845.7	33.70	34.62	35.33	26.53	27.07	27.37	845.7	875.7	15.62	15.64	15.63
905.4	32.94	33.74	34.36	25.59	25.92	26.07	905.4	935.4	14.66	14.61	14.58
945.2	32.65	33.39	33.90	25.15	25.36	25.35	945.2	975.2	14.18	14.11	14.07
1004.8	32.39	33.13	33.57	24.57	24.64	24.35	1004.8	1034.8	13.67	13.60	13.54
1044.6	32.16	32.91	33.36	24.12	24.17	23.78	1044.6	1074.6	13.44	13.37	13.33
1104.3	31.89	32.59	33.03	23.70	23.66	23.09	1104.3	1134.3	13.21	13.16	13.14
1144.1	31.75	32.41	32.84	23.44	23.43	22.85	1144.1	1174.1	13.06	13.02	12.98
1203.8	31.60	32.15	32.52	23.00	23.05	22.62	1203.8	1233.8	12.89	12.84	12.81
1243.6	31.58	32.03	32.35	22.53	22.58	22.33	1243.6	1273.6	12.75	12.69	12.67
1303.3	31.59	31.90	32.06	21.45	21.18	20.92	1303.3	1333.3	12.56	12.43	12.33
1343.0	31.54	31.72	31.81	20.63	20.08	19.65	1343.0	1373.0	12.35	12.16	12.02
1402.7	31.40	31.41	31.44	19.80	19.00	18.49	1402.7	1432.7	11.86	11.63	11.47
1442.5	31.29	31.16	31.14	19.40	18.56	18.04	1442.5	1472.5	11.46	11.21	11.03
1502.2	31.18	30.82	30.62	19.01	18.10	17.49	1502.2	1532.2	10.86	10.57	10.36
1542.0	31.23	30.67	30.33	18.96	17.97	17.30	1542.0	1572.0	10.38	10.08	9.86
1601.7	30.92	29.97	29.37	18.78	17.63	16.91	1601.7	1631.7	9.70	9.37	9.14
1641.5	30.61	29.34	28.56	18.67	17.39	16.57	1641.5	1671.5	9.22	8.87	8.65
1701.2	30.08	28.40	27.40	18.55	17.18	16.28	1701.2	1731.2	8.54	8.15	7.87
1740.9	29.78	27.83	26.70	18.45	17.03	16.09	1740.9	1770.9	8.13	7.73	7.48
1800.6	29.22	26.97	25.70	18.31	16.86	15.89	1800.6	1830.6	7.51	7.11	6.86
1840.4	28.86	26.56	25.26	18.20	16.76	15.78	1840.4	1870.4	7.17	6.78	6.53
1900.1	28.30	25.88	24.49	18.09	16.67	15.65	1900.1	1930.1	6.56	6.20	5.98

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Typical Performance Data

RF (IN) (MHz)	LO (MHz)	RF VSWR (:1)			LO (MHz)	LO VSWR (:1)			IF (OUT) (MHz)	IF VSWR @LO=1500.1MHz (:1)		
		@LO (dBm)				@LO (dBm)				@LO (dBm)		
		+4	+7	+10		+4	+7	+10		+4	+7	+10
5.0	35.0	1.68	1.59	1.54	5.0	1.11	1.91	2.80	0.1	1.32	1.10	1.02
10.0	40.0	1.64	1.51	1.45	10.0	1.11	1.94	2.76	5.0	1.39	1.16	1.06
49.9	79.9	1.41	1.30	1.24	49.9	1.12	1.69	2.45	10.0	1.39	1.15	1.05
89.7	119.7	1.49	1.38	1.32	89.7	1.12	1.69	2.42	30.3	1.80	1.54	1.43
129.5	159.5	1.42	1.33	1.28	129.5	1.14	1.68	2.39	50.5	1.64	1.44	1.24
169.3	199.3	1.47	1.38	1.32	169.3	1.17	1.71	2.43	70.7	1.59	1.38	1.25
209.0	239.0	1.44	1.35	1.32	209.0	1.17	1.69	2.37	90.9	1.69	1.47	1.32
248.8	278.8	1.41	1.34	1.30	248.8	1.21	1.73	2.44	111.1	1.74	1.50	1.37
308.5	338.5	1.43	1.37	1.34	308.5	1.22	1.75	2.46	131.3	1.68	1.46	1.33
348.3	378.3	1.41	1.35	1.32	348.3	1.25	1.82	2.55	151.5	1.63	1.42	1.30
408.0	438.0	1.40	1.35	1.33	408.0	1.26	1.83	2.55	171.7	1.66	1.43	1.31
447.8	477.8	1.45	1.40	1.37	447.8	1.30	1.88	2.63	191.9	1.71	1.48	1.36
507.5	537.5	1.48	1.43	1.40	507.5	1.32	1.92	2.67	212.1	1.70	1.48	1.35
547.3	577.3	1.47	1.43	1.41	547.3	1.36	1.99	2.75	232.3	1.65	1.43	1.31
606.9	636.9	1.49	1.45	1.43	606.9	1.39	2.01	2.76	252.5	1.61	1.41	1.29
646.7	676.7	1.43	1.40	1.39	646.7	1.45	2.09	2.87	272.8	1.61	1.42	1.30
706.4	736.4	1.42	1.40	1.39	706.4	1.49	2.12	2.88	293.0	1.64	1.44	1.32
746.2	776.2	1.33	1.31	1.30	746.2	1.54	2.20	2.99	313.2	1.63	1.43	1.31
805.9	835.9	1.30	1.28	1.28	805.9	1.58	2.22	2.98	333.4	1.58	1.39	1.29
845.7	875.7	1.27	1.23	1.22	845.7	1.64	2.29	3.06	353.6	1.56	1.38	1.29
905.4	935.4	1.24	1.20	1.19	905.4	1.71	2.37	3.14	394.0	1.59	1.42	1.33
945.2	975.2	1.26	1.21	1.19	945.2	1.79	2.46	3.23	434.4	1.55	1.39	1.31
1004.8	1034.8	1.16	1.12	1.12	1004.8	1.88	2.57	3.35	454.6	1.55	1.40	1.33
1044.6	1074.6	1.15	1.11	1.11	1044.6	1.95	2.64	3.43	495.0	1.57	1.43	1.36
1104.3	1134.3	1.06	1.05	1.10	1104.3	2.03	2.73	3.52	515.2	1.55	1.42	1.36
1144.1	1174.1	1.08	1.09	1.13	1144.1	2.10	2.81	3.61	555.6	1.56	1.44	1.39
1203.8	1233.8	1.25	1.25	1.26	1203.8	2.15	2.84	3.62	575.8	1.57	1.46	1.41
1243.6	1273.6	1.29	1.30	1.32	1243.6	2.23	2.93	3.73	616.2	1.56	1.47	1.43
1303.3	1333.3	1.37	1.37	1.38	1303.3	2.22	2.86	3.62	636.4	1.57	1.48	1.45
1343.0	1373.0	1.44	1.45	1.47	1343.0	2.29	2.94	3.70	676.8	1.59	1.51	1.48
1402.7	1432.7	1.50	1.52	1.54	1402.7	2.29	2.90	3.63	697.0	1.57	1.50	1.47
1442.5	1472.5	1.60	1.63	1.64	1442.5	2.36	2.97	3.71	737.4	1.58	1.53	1.52
1502.2	1532.2	1.80	1.83	1.84	1502.2	2.44	3.03	3.76	757.7	1.60	1.55	1.53
1542.0	1572.0	1.90	1.93	1.94	1542.0	2.55	3.12	3.83	798.1	1.56	1.51	1.50
1601.7	1631.7	2.08	2.11	2.12	1601.7	2.71	3.26	3.94	818.3	1.55	1.52	1.52
1641.5	1671.5	1.99	2.03	2.04	1641.5	2.84	3.35	4.01	858.7	1.58	1.56	1.55
1701.2	1731.2	2.02	2.05	2.06	1701.2	3.03	3.48	4.10	878.9	1.56	1.53	1.53
1740.9	1770.9	1.88	1.91	1.93	1740.9	3.20	3.62	4.21	919.3	1.51	1.50	1.51
1800.6	1830.6	1.92	1.93	1.94	1800.6	3.36	3.70	4.21	939.5	1.52	1.51	1.53
1840.4	1870.4	2.03	2.04	2.07	1840.4	3.56	3.84	4.33	979.9	1.49	1.47	1.48
1900.1	1930.1	2.09	2.09	2.10	1900.1	3.69	3.88	4.29	1000.1	1.44	1.44	1.45

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Harmonics Tables

RF HARMONICS ORDER

	(-dBm)	(-dBc)										
0	-	-	+0	20	8	36	18	27	34	38	58	44
1	-	10	+0	27	14	39	33	41	47	38	59	48
2	124	63	48	59	50	59	51	62	63	61	69	64
3	117	68	79	67	67	68	65	86	79	77	81	82
4	118	96	90	91	83	81	83	91	90	92	92	94
5	116	107	103	130	110	95	86	104	100	99	102	105
6	119	102	109	100	99	119	104	83	119	102	114	112
7	114	101	103	114	115	125	100	95	102	109	109	109
8	109	99	103	114	110	97	110	101	97	91	101	106
9	117	95	106	112	107	106	102	109	108	95	97	98
10	114	99	98	102	122	111	107	102	104	110	104	96
RF CAL	0	1	2	3	4	5	6	7	8	9	10	

LO HARMONICS ORDER

Test conditions:

RF IN: 750.1 MHz; -14.00 dBm.

LO IN: 780.01 MHz; +7.00 dBm

IF OUT: 29.91 MHz; -21.11 dBm

RF HARMONICS ORDER

	(-dBm)	(-dBc)										
0	-	-	10	31	18	42	28	39	48	53	63	66
1	-	10	+0	28	14	42	33	46	56	43	67	54
2	89	51	37	47	39	48	42	57	71	54	67	60
3	116	47	60	49	50	50	47	65	61	60	63	59
4	115	78	71	69	58	61	58	77	61	70	64	64
5	114	73	76	73	66	66	62	72	64	73	71	75
6	111	91	87	91	88	78	70	71	70	72	72	78
7	122	92	97	94	107	87	78	78	78	79	85	89
8	106	110	104	99	102	116	92	91	83	82	82	93
9	113	121	111	116	109	104	105	103	107	105	94	97
10	105	111	113	111	130	111	114	116	106	99	99	98
RF CAL	0	1	2	3	4	5	6	7	8	9	10	

LO HARMONICS ORDER

Test conditions:

RF IN: 750.1 MHz; -4.00 dBm.

LO IN: 780.01 MHz; +7.00 dBm

IF OUT: 29.91 MHz; -11.16 dBm

Notes: 1. All Harmonics are in (dBc) relative to IF OUTPUT.

2. + entry denotes harmonics are in (dBc) above IF OUTPUT.

3. RF Cal represent the Harmonics level of the RF input signal to the mixer.

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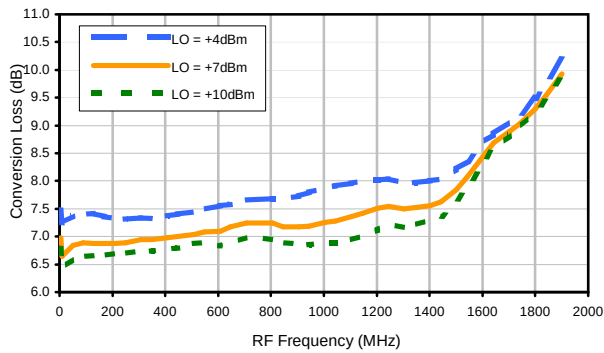


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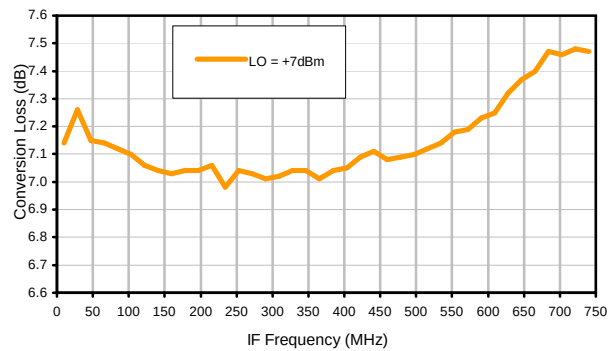


Typical Performance Curves

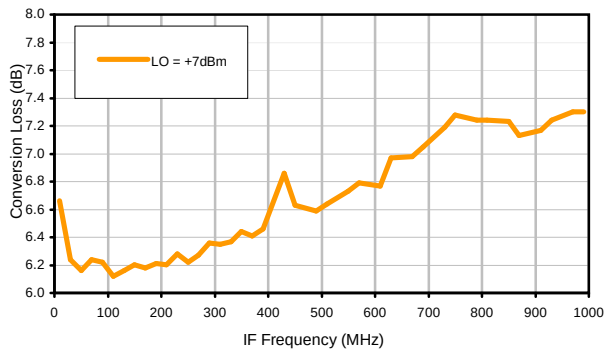
Conversion Loss @ IF=30MHz



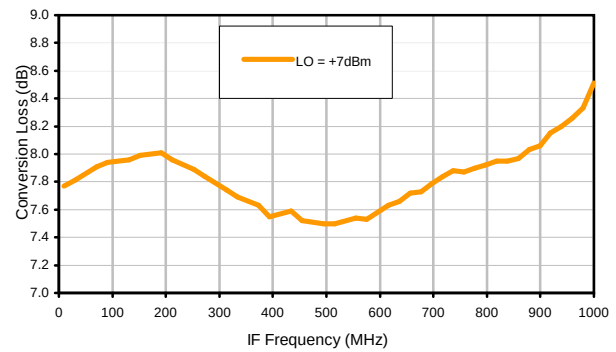
Conversion Loss vs. IF @ RF=750.1MHz



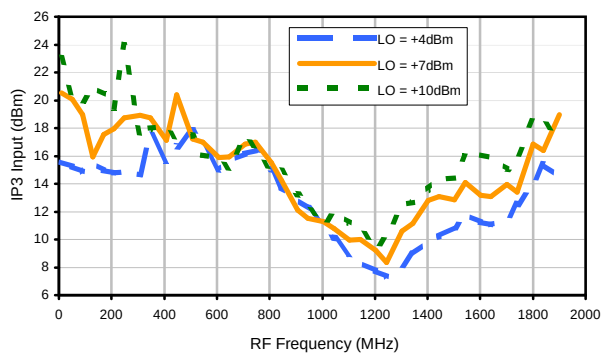
Conversion Loss vs. IF @ RF=10.1MHz



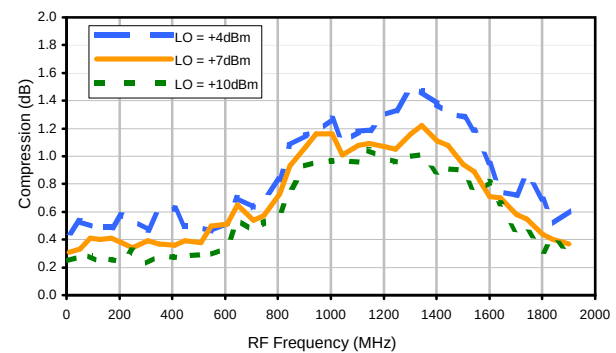
Conversion Loss vs. IF @ RF=1500.1MHz



IP3 Input

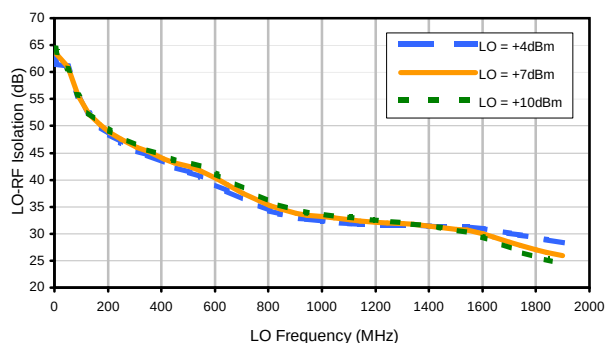


Compression @ RF IN=+1dBm

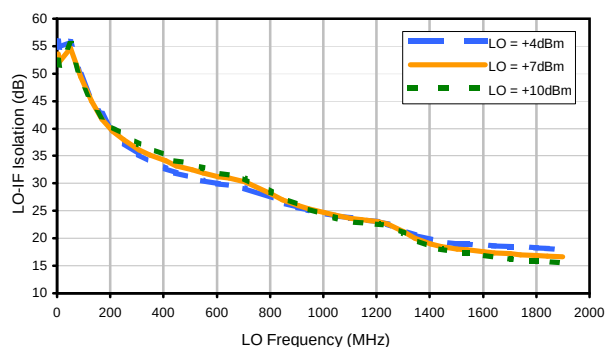


Typical Performance Curves

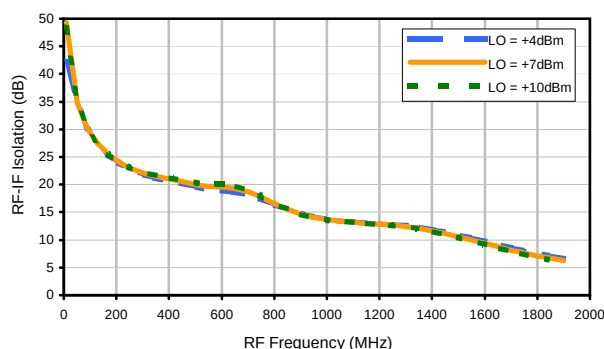
LO-RF Isolation



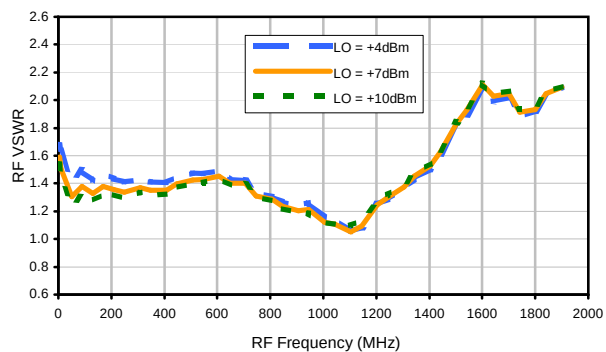
LO-IF Isolation



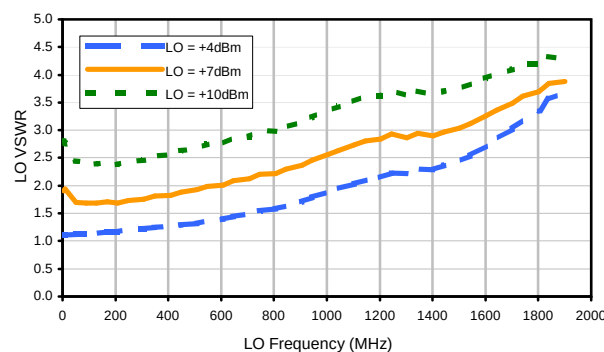
RF-IF Isolation



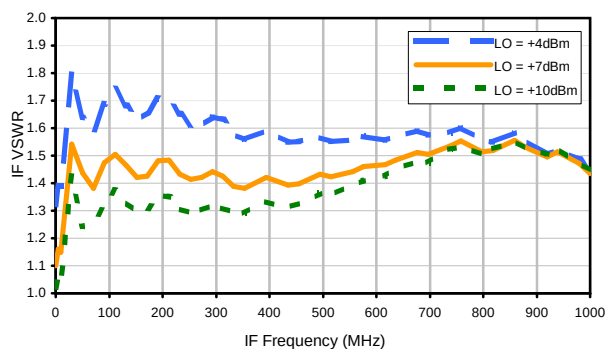
RF VSWR



LO VSWR



IF VSWR



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Harmonics Tables

RF HARMONICS ORDER

	(-dBm)	(-dBc)										
0	-	-	+0	20	8	36	18	27	34	38	58	44
1	-	10	+0	27	14	39	33	41	47	38	59	48
2	124	63	48	59	50	59	51	62	63	61	69	64
3	117	68	79	67	67	68	65	86	79	77	81	82
4	118	96	90	91	83	81	83	91	90	92	92	94
5	116	107	103	130	110	95	86	104	100	99	102	105
6	119	102	109	100	99	119	104	83	119	102	114	112
7	114	101	103	114	115	125	100	95	102	109	109	109
8	109	99	103	114	110	97	110	101	97	91	101	106
9	117	95	106	112	107	106	102	109	108	95	97	98
10	114	99	98	102	122	111	107	102	104	110	104	96
RF CAL	0	1	2	3	4	5	6	7	8	9	10	

LO HARMONICS ORDER

Test conditions:

RF IN: 750.1 MHz; -14.00 dBm.
LO IN: 780.01 MHz; +7.00 dBm
IF OUT: 29.91 MHz; -21.11 dBm

RF HARMONICS ORDER

	(-dBm)	(-dBc)										
0	-	-	10	31	18	42	28	39	48	53	63	66
1	-	10	+0	28	14	42	33	46	56	43	67	54
2	89	51	37	47	39	48	42	57	71	54	67	60
3	116	47	60	49	50	50	47	65	61	60	63	59
4	115	78	71	69	58	61	58	77	61	70	64	64
5	114	73	76	73	66	66	62	72	64	73	71	75
6	111	91	87	91	88	78	70	71	70	72	72	78
7	122	92	97	94	107	87	78	78	78	79	85	89
8	106	110	104	99	102	116	92	91	83	82	82	93
9	113	121	111	116	109	104	105	103	107	105	94	97
10	105	111	113	111	130	111	114	116	106	99	99	98
RF CAL	0	1	2	3	4	5	6	7	8	9	10	

LO HARMONICS ORDER

Test conditions:

RF IN: 750.1 MHz; -4.00 dBm.
LO IN: 780.01 MHz; +7.00 dBm
IF OUT: 29.91 MHz; -11.16 dBm

Notes: 1. All Harmonics are in (dBc) relative to IF OUTPUT.

2. + entry denotes harmonics are in (dBc) above IF OUTPUT.

3. RF Cal represent the Harmonics level of the RF input signal to the mixer.

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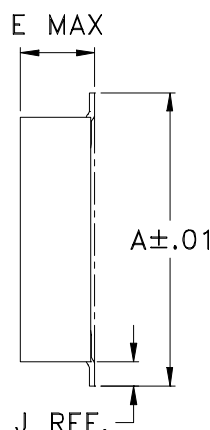
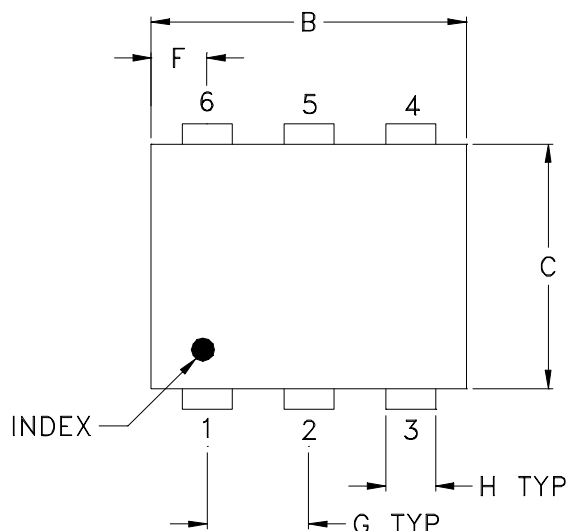


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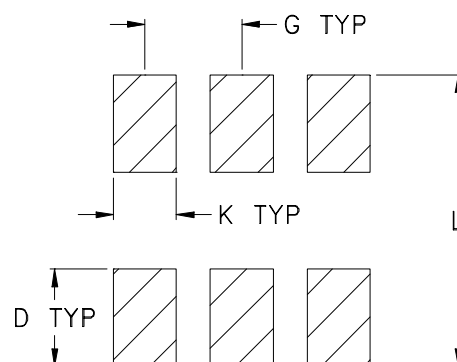


CD541
CD542
CD636
CD637

Outline Dimensions



PCB Land Pattern



Suggested Layout,
Tolerance to be within $\pm .002$

CASE#	A	B	C	D	E	F	G	H	J	K	L	WT, GRAM
CD541					.082 (2.08)							.15
CD542	.272 (6.91)	.310 (7.87)	.220 (5.58)	.100 (2.54)	.112 (2.84)	.055 (1.40)	.100 (2.54)	.030 (0.76)	.026 (0.66)	.065 (1.65)	.300 (7.62)	.20
CD636					.162 (4.11)							.25
CD637					.206 (5.23)							.40

Dimensions are in inches (mm). Tolerances: 2 Pl. $\pm .01$; 3 Pl. $\pm .005$

Notes:

- Case material: Plastic.
- Termination finish:
For RoHS Case Styles: Tin plate over Nickel plate. All models, (+) suffix.
For RoHS-5 Case Styles: Tin-Lead plate. All models, no (+) suffix.



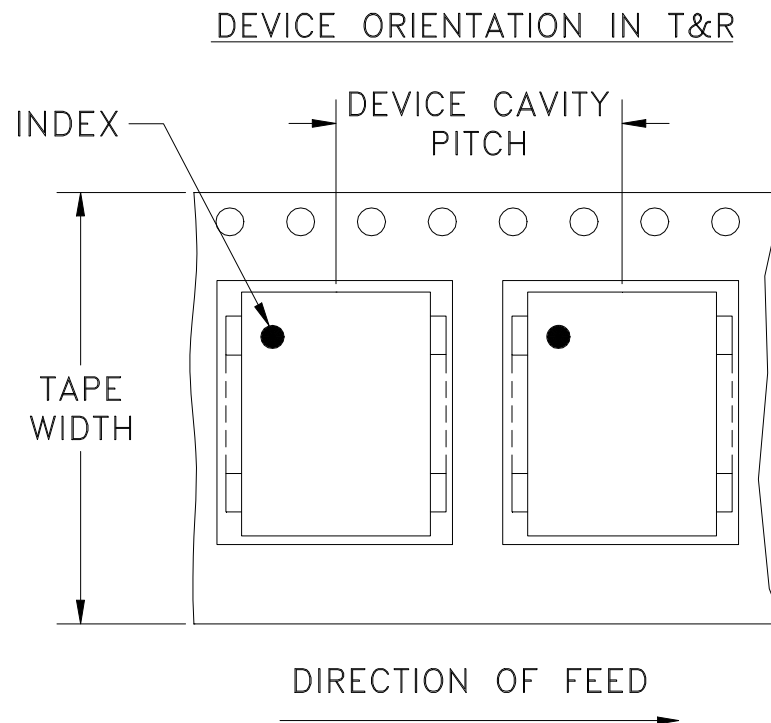
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Tape Width, mm	Device Cavity Pitch, mm	Reel Size, inches	Devices per Reel see note	
16	12	7	Small quantity standard (see note)	20
				50
				100
				200
		13	Standard	500
				1000

Note: Availability of small reel quantity varies by model.
Refer to pricing and availability on individual model dashboard.

Mini-Circuits carrier tape materials provide protection from ESD (Electro-Static Discharge) during handling and transportation. Tapes are static dissipative and comply with industry standards EIA-481/EIA-541.

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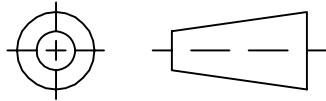
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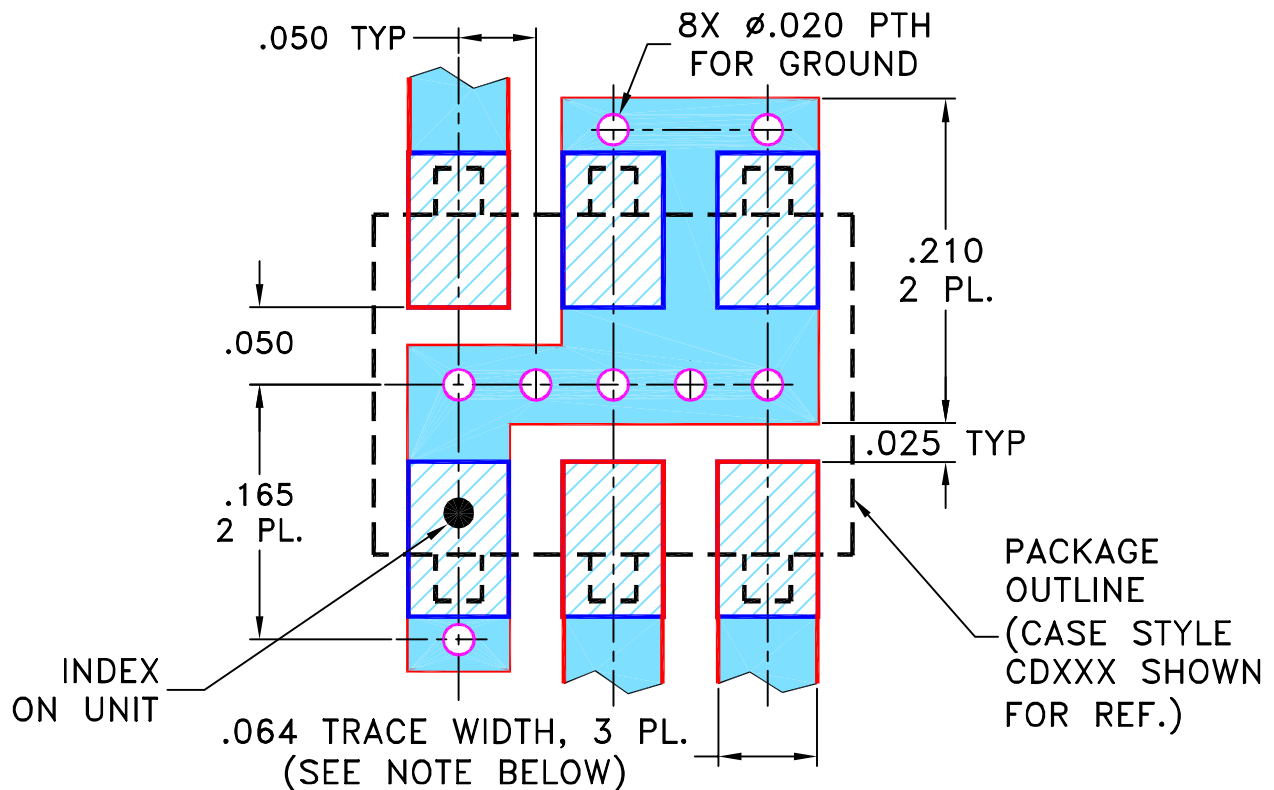
THIRD ANGLE PROJECTION



REVISIONS

REV	ECN No.	DESCRIPTION	DATE	DR	AUTH
A	M101143	ADDED "gk" PIN CONNECTION, TT100 CASE STYLE & NOTE 2	10/10/05	MMG	DJ
B	M102713	ADDED "...WITH SMOBC"	01/17/06	MMG	IL
C	M108637	REMOVED "PIN 1", ADDED INDEX ON UNIT	12/01/06	MYG	FL

SUGGESTED MOUNTING CONFIGURATION
FOR BH292, CD541/542/636/637, TT100/240 CASE
STYLES, "gk", "ht", "hu", "nd", "w" PIN CONNECTIONS



- NOTES:** 1. TRACE WIDTH IS SHOWN FOR ROGERS R04350B WITH DIELECTRIC THICKNESS .030" $\pm$.002"; COPPER: 1/2 OZ. EACH SIDE.
 FOR OTHER MATERIALS TRACE WIDTH MAY NEED TO BE MODIFIED.
 2. BOTTOM SIDE OF THE PCB IS CONTINUOUS GROUND PLANE.



DENOTES PCB COPPER LAYOUT WITH SMOBC
 (SOLDER MASK OVER BARE COPPER)



DENOTES COPPER LAND PATTERN FREE OF SOLDER MASK

UNLESS OTHERWISE SPECIFIED

INITIALS

DATE

DIMENSIONS ARE IN INCHES

TOLERANCES ON:

2 PL DECIMALS $\pm$ 3 PL DECIMALS $\pm$.005ANGLES $\pm$ FRACTIONS $\pm$

DRAWN

MMG

07/17/02

CHECKED

WL

08/02/02

APPROVED

DJ

08/05/02



Mini-Circuits®

13 Neptune Avenue
 Brooklyn NY 11235

PL, gk/ht/hu/nd/w, BH292,
 CD541/542/636/637, TT100/240, TB-03

SIZE

A

CODE IDENT

15542

DRAWING NO:

98-PL-052

REV:

C

FILE:

98PL052

SCALE:

8:1

SHEET:

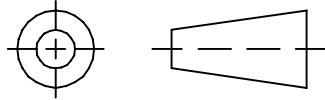
1 OF 1

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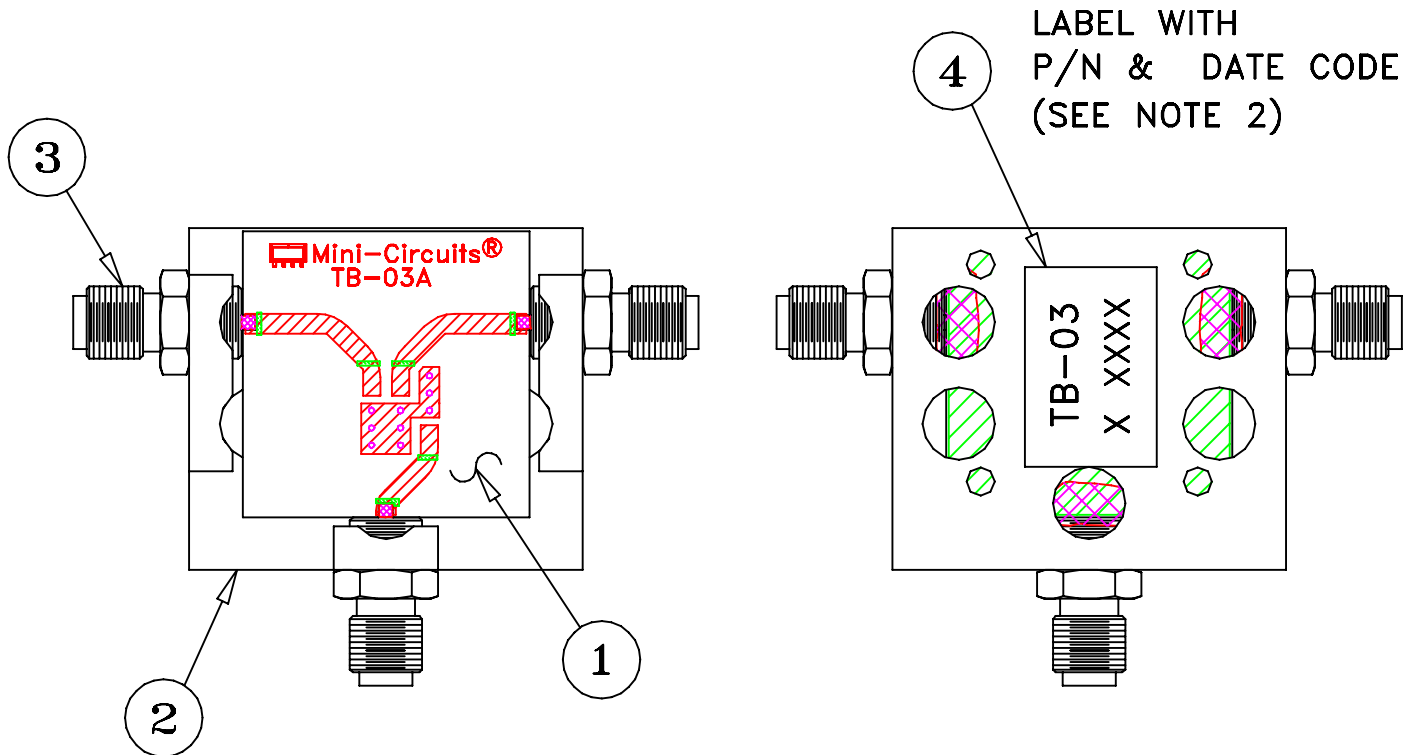
ASHEETA1.DWG REV:A DATE:01/12/95

THIRD ANGLE PROJECTION



REVISIONS

REV	ECN No.	DESCRIPTION	DATE	DR	AUTH
E	M119737	UPDATED PCB	10.08	MF	AD
F	M127659	UPDATED CARR	06.10	SW	SG
G	M127846	UPDATED SCHEMATIC DIAGRAM	06.10	SW	SG
H	M131840	UPDATED DWG	05.11	MF	AD

**NOTES:**

1. REFER TO -09 PAGE FOR ITEM DESCRIPTIONS.
DESIGNATION NUMBERS ON -20 PAGE CORRESPOND TO THE NUMBERS ON -09 PAGE.
2. FOR TEXT HEIGHT & STYLE ON THE LABEL REFER TO: D3-G209.

UNLESS OTHERWISE SPECIFIED

INITIALS

DATE

DIMENSIONS ARE IN INCHES

TOLERANCES ON:
2 PL DECIMALS ±
3 PL DECIMALS ±
ANGLES ±
FRACTIONS ±



DRAWN

S.WOLYNSKI

06.29.99

CHECKED

SG

07.06.99

APPROVED

MG

07.10.99



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13 Neptune Avenue
Brooklyn NY 11235

TB,ADE,CD542/636,06MX01,50

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ASHEETA1.DWG REV:A DATE:01/12/95

SIZE

A

CODE IDENT

15542

DRAWING NO:

TB-03-20

REV:

H

FILE:

WTB-03

SCALE:

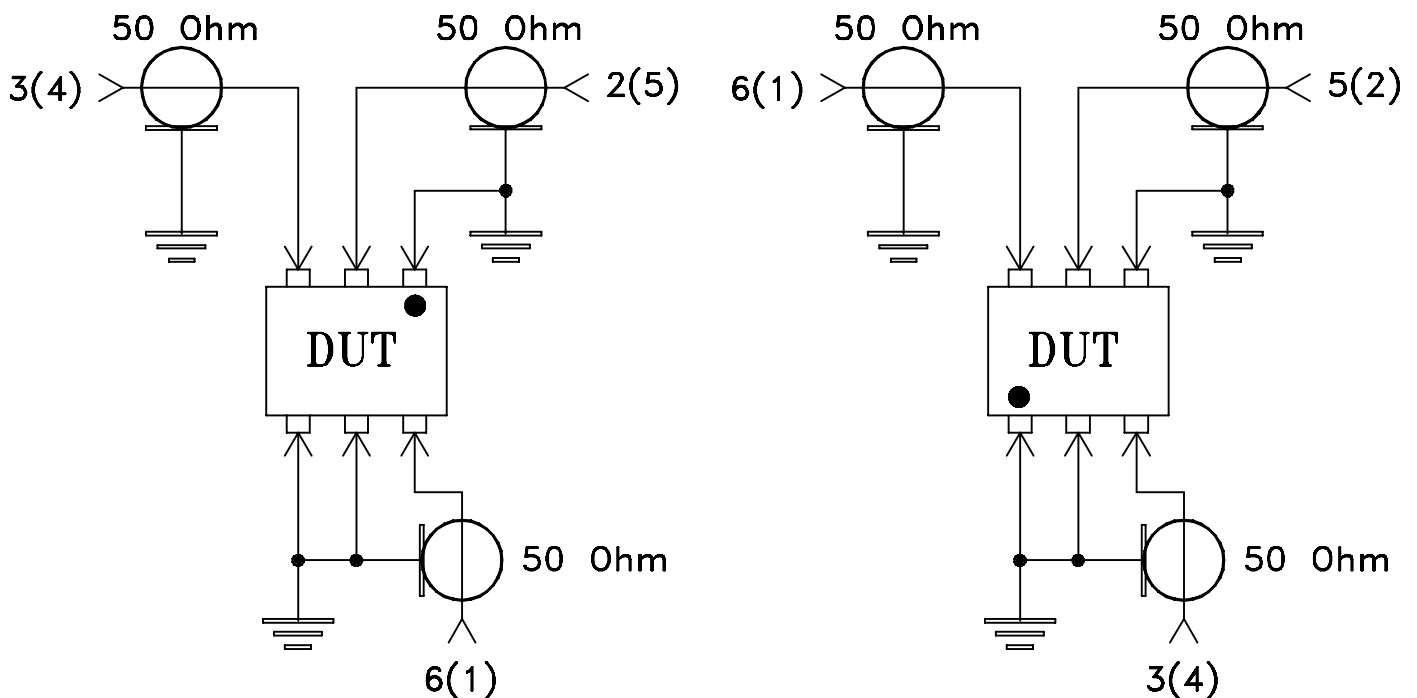
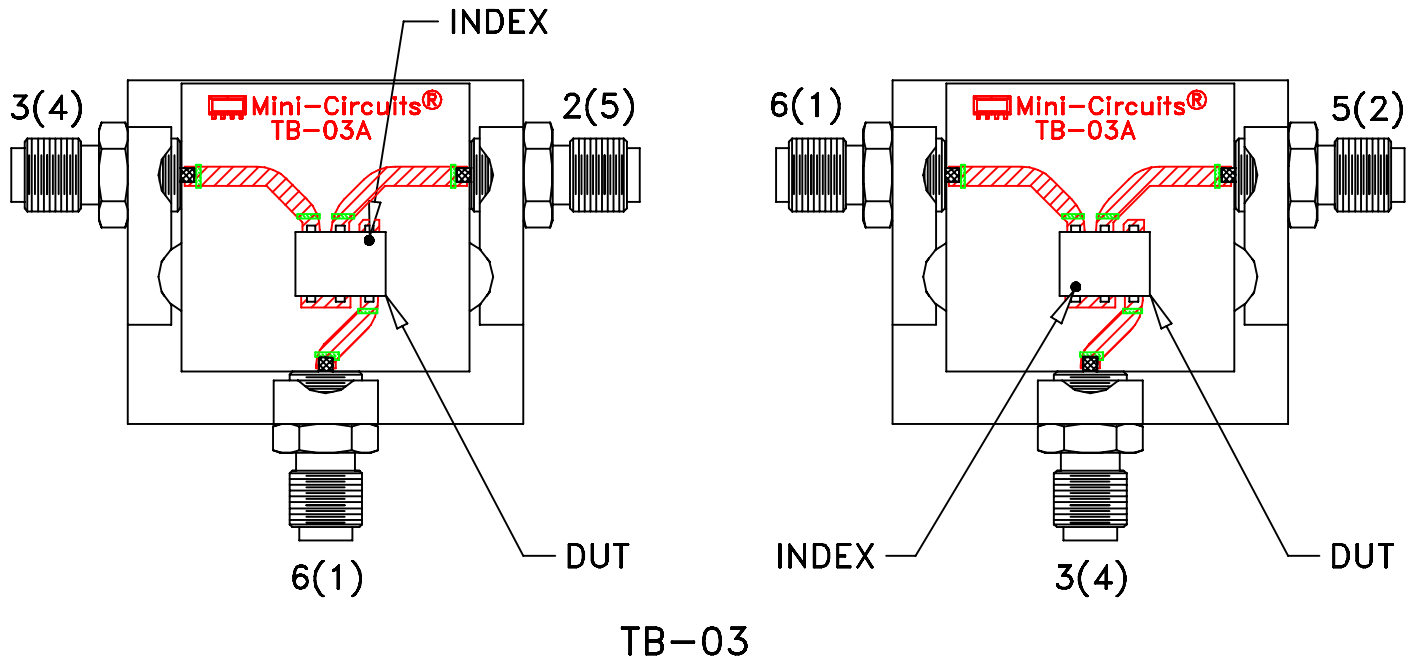
1.5:1

SHEET:

1 OF 2

Evaluation Board and Circuit

For Pin Connections and DUT Orientation Refer to
Data Sheet of the DUT



Schematic Diagram

Notes:

1. 50 Ohm SMA Female connectors.
2. PCB Material: Rogers R04350 or equivalent,
Dielectric Constant=3.5, Thickness=.030 inch.

 **Mini-Circuits®**



All Mini-Circuits products are manufactured under exacting quality assurance and control standards, and are capable of meeting published specifications after being subjected to any or all of the following physical and environmental test.

Specification	Test/Inspection Condition	Reference/Spec
Operating Temperature	-40° to 85°C Ambient Environment	Individual Model Data Sheet
Storage Temperature	-55° to 100° C Ambient Environment	Individual Model Data Sheet
Humidity	90 to 95% RH, 240 hours, 50°C	MIL-STD-202, Method 103, Condition A, Except 50°C and end-point electrical test done within 12 hours
Thermal Shock	-55° to 100°C, 100 cycles	MIL-STD-202, Method 107, Condition A-3, except +100°C
Solder Reflow Heat	Sn-Pb Eutetic Process: 225°C peak Pb-Free Process 245° - 250°C peak	J-STD-020, Table 4-1, 4-2 and 5-2, Figure 5-1
Solderability	10X Magnification	J-STD-002, 95% Coverage
Vibration (High Frequency)	20g peak, 10-2000 Hz, 12 times in each of three perpendicular directions (total 36)	MIL-STD-202, Method 204, Condition D
Mechanical Shock	50g, 11 ms, 1/2-sine, 18 shocks: 3 each direction, each of 3 axes	MIL-STD-202, Method 213, Condition A
Marking Resistance to Solvents	Isopropyl alcohol + mineral spirits at 25°C; terpene defluxer at 25°C; distilled water + proylene glycol monomethyl ether + monoethanolamine at 63°C to 70°C	MIL-STD-202, Method 215